Lijie Yao 姚李捷

Ph.D. M.Sc. Ing. B.Eng.

Business Address: 111 Renai Rd, Xi'an Jiaotong-Liverpool University, SD547, Suzhou, Jiangsu, China, 215123
Contact: yaolijie0219@gmail.com | lijieyao.com
Languages Spoken: Chinese (native fluency) | English (advanced proficiency) | French (advanced proficiency)

ABOUT ME

I am an **Assistant Professor** and **Program Director of Computer Science and Technology** at **Xi'an Jiaotong-Liverpool University (XJTLU)**. I obtained my Ph.D. degree from **Université Paris-Saclay (UPS)**. My work received the **Ph.D. Thesis Prize** from **ACM AFIHM** and **Canada Mitacs Globalink Research Award**. Part of my work on *visualization in motion* was applied in the **2024 Paris Olympics** swimming races. Previously, I was a postdoctoral and doctoral researcher at **INRIA** and a visiting researcher at **University of Calgary**. My research interests include *visualization in motion*, *AI-assisted data communication*, and *human-AI interaction*. Collaboration emails are more than welcome.

EDUCATION

Université Paris-Saclay Gif-sur-Yvette, France Sept. 2020 – Dec. 2023
Doctor of Philosophy – Computer Science

- Supervisors: Petra Isenberg & Anastasia Bezerianos
- Thesis Topic: Situated Visualization in Motion
- Research Focus: Visualization in motion, perception, visual design

Université Paris-Saclay Gif-sur-Yvette, France Sept. 2019 – Aug. 2020
Master of Science – Computer Science

- Supervisors: Petra Isenberg
- Thesis Topic: Situated Visualization in Motion
- Research Focus: Visualization in motion, perception, visual design

Polytech Lille Lille, France Sept. 2017 – Aug. 2019
Diplôme d'ingénieur (= Master of Engineering) – Electrical Engineering and Automation

- Supervisors: Komi Midzodzi Pekpe
- Thesis Topic: Position Control of A Drone
- Research Focus: Embedded systems, automatic control

XIDIAN University | 西安电子科技大学 Xi'an, China Aug. 2014 – June 2018
Bachelor of Engineering – Electronic Information Engineering

- Supervisors: Xavier Redon & Alexandre Boé
- Thesis Topic: Electronic Necklace for Health Monitoring
- Research Focus: Embedded systems, microelectronics

WORK EXPERIENCE

Assistant Professor Xi'an Jiaotong-Liverpool University July 2024 – Now
Suzhou, Jiangsu, China

- Xi'an Jiaotong-Liverpool University is a Chinese private university founded in 2006 as a joint venture between the University of Liverpool and Xi'an Jiaotong University.
- My duties include research, teaching, and academic/administrative services.

Postdoctoral Researcher Inria Jan. 2024 – March 2024

Gif-Sur-Yevette, Île-de-France, France

- Inria is a French national institution for research in digital science and technology. It is ranked 18th in the global research institution and 7th in computer science.
- My duties included reviewing literature, conducting experiments, and writing papers.

Visiting Researcher University of Calgary July 2023 – Oct. 2023

Calgary, Alberta, Canada

- The University of Calgary is a Canadian public research university founded in 1908, prior to being instituted into a separate, autonomous university in 1966.
- My duties included reviewing literature and conducting experiments.

Teaching Assistant Université Paris-Saclay Sept. 2021 – Aug. 2023

Gif-Sur-Yevette, Île-de-France, France

- The Université Paris-Saclay is a french combined technological research institute and public research university. It is ranked 13th in the ARWU 2025 and 70th in QS 2026.
- My duties included reviewing literature and conducting experiments.

Doctoral Researcher Inria Sept. 2020 – Dec. 2023

Gif-Sur-Yevette, Île-de-France, France

- Inria is a French national institution for research in digital science and technology. It is ranked 18th in the global research institution and 7th in computer science.
- My duties included reviewing literature, conducting experiments, writing papers, advising students, and presenting.

PEER-REVIEWED PUBLICATIONS

* Corresponding author, † These authors contribute equally. List order: reverse chronology.

Journals

1. Junhao Zhao, Romain Vuillemot, Petra Isenberg, **Lijie Yao***. Authoring Narrative Visualization in Motion: Visual Storytelling in Swimming Videos. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2027. In press.
2. Junxiu Tang, **Lijie Yao***, Lu Ying, Romain Vuillemot, Petra Isenberg. Diving Deep into Time: Temporal Arrangements for Embedded Visualization in Swimming Videos. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2026, 32(7), pp.6825-6841. DOI: 10.1109/TVCG.2026.3689361.
3. **Lijie Yao***, Federica Bucchieri, Victoria McArthur, Anastasia Bezerianos, Petra Isenberg. User Experience of Visualizations in Motion: A Case Study and Design Considerations. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025, 31 (1), pp.174-184. DOI: 1109/TVCG.2024.3456319.
4. **Lijie Yao***. Situated Visualization in Motion. *IEEE Computer Graphics and Applications (CG&A)*. 2024, 44(6), pp.142-150. DOI: 10.1109/MCG.2024.3462129.
5. **Lijie Yao***, Romain Vuillemot, Anastasia Bezerianos, Petra Isenberg. Designing for Visualization in Motion: Embedding Visualizations in Swimming Videos. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2024, 30 (3), pp.1821-1836. DOI: 10.1109/TVCG.2023.3341990.
6. **Lijie Yao***, Anastasia Bezerianos, Romain Vuillemot, Petra Isenberg. Visualization in Motion: A Research Agenda and Two Evaluations. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2022, 28(10), pp.3546-3562, DOI:

10.1109/TVCG.2022.3184993.

Conferences

7. Best Paper Honorable Mention Jingyu Wang, Jiahan Zhang, Yu Liu, Lingyun Yu, **Lijie Yao***. Tangible Carbon Blocks: Data Physicalization for Everyday Carbon Emissions. *ChinaVis*, July 2026, Guiyang, China.
8. Zehong Ye, Jingyu Wang, Xiao Xie, Lingyun Yu, **Lijie Yao***. When Color Contrast Is Not Enough: An Empirical Study of Dynamic Color Adaptation for In-game Health Visualizations. *ChinaVis*, July 2026, Guiyang, China.
9. Wanfang Xu, Jifan Yang, Fengwen Zhang, Yu Lu, **Lijie Yao**, Le Liu, Lingyun Yu*. Touch, Sound, and Space: Exploring Immersive Music Interaction through AI-Generated Environments. *International Symposium on Visual Information Communication and Interaction (VINCI)*, Dec. 2025, Linz, Austria, Article 23, pp. 1-8. DOI: 10.1145/3769534.3769589.
10. Haonan Yao, Lingyun Yu, **Lijie Yao***. Designing Visualization Widgets for Tangible Data Exploration: A Systematic Review. *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 2025, Vienna, Austria, pp. 261-265. DOI: 10.1109/VIS60296.2025.00058.
11. **Lijie Yao***, Yuming Zhang, Yu Liu. Input Physicalization in Practice: An Instructional Walkthrough with Visualization Novices. *IEEE VIS workshop on Input Visualization*, Nov. 2025, Vienna, Austria. Link: osf.io/dfgvx
12. Qi Guo†, Junyi Li†, Jiayi Hong, **Lijie Yao***. Early Exploration into AI-Assisted Visual Analytics for Dynamic Videos. *IEEE VIS workshop on GenAI, Agents, and the Future of VIS*, Nov. 2025, Vienna, Austria. Link: visxgenai.github.io/subs-2025/1558/1558-doc.pdf
13. Junhao Zhao, **Lijie Yao***. Structured AI Agents for Reliable Visualization Report Generation. *IEEE VIS workshop on GenAI, Agents, and the Future of VIS*, Nov. 2025, Vienna, Austria. Link: nemozjh.github.io/VIS_Agent_nemo/
14. Best Poster Honorable Mention **Lijie Yao***. Visualization in Motion: Perception, Design, and User Experience. *Posters of ChinaVis*, July 2025, Hangzhou, China.
15. Fairouz Grioui*, Tanja Blascheck, **Lijie Yao**, Petra Isenberg. Micro Visualizations on a Smartwatch: Assessing Reading Performance While Walking. *IEEE Visualization & Visual Analytics (IEEE VIS)*, Oct. 2024, Florida, U.S. DOI: 10.1109/VIS55277.2024.00017.
16. Ahmed Elshabasi*, **Lijie Yao**, Petra Isenberg, Charles Perin, Wesley Willett. Collecting Information Needs for Egocentric Visualizations while Running. *IEEE VIS workshop on First-Person Visualizations for Outdoor Physical Activities*, Oct. 2024, Florida, U.S. DOI: 10.48550/arXiv.2409.06426.
17. Yvonne Jansen, Federica Bucchieri, Pierre Dragicevic, Martin Hachet, Morgane Koval, Léana Petiot, Arnaud Prouzeau, Dieter Schmalstieg, **Lijie Yao**, Petra Isenberg*. Envisioning Situated Visualizations of Environmental Footprints in an Urban Environment. *IEEE VIS workshop on Visualization for Social Good*, Oct. 2022, Oklahoma City, U.S. DOI: 10.5281/zenodo.7053934.
18. Alaul Islam†, **Lijie Yao***†, Anastasia Bezerianos, Tanja Blascheck, Tingying He, et al.. Reflections on Visualization in Motion for Fitness Trackers. *ACM MobileHCI workshop on New Trends in HCI and Sports*, Sep. 2022, Vancouver, Canada. DOI: 10.48550/arXiv.2409.06401.
19. Federica Bucchieri*, **Lijie Yao**, Petra Isenberg. Situated Visualization in Motion for Video Games. *Posters of the EuroVis*, Jun 2022, Rome, Italy. DOI: 10.2312/evp.20221119.
20. **Lijie Yao***, Anastasia Bezerianos, Romain Vuillemot, Petra Isenberg. Situated Visualization in Motion for Swimming. *Journée Visu 2022*, Jun 2022, Bordeaux, France. DOI: 10.48550/arXiv.2409.07695.
21. Federica Bucchieri*, **Lijie Yao**, Petra Isenberg. Visualization in Motion in Video Games for Different Types of Data. *Journée-Visu 2022*, Jun 2022, Bordeaux, France. DOI: 10.48550/arXiv.2409.07696.
22. **Lijie Yao***, Anastasia Bezerianos, Petra Isenberg. Situated Visualization in Motion. *Posters of the IEEE VIS*, Oct 2020, Salt

Lake City, U.S.. 2020. DOI: 10.48550/arXiv.2409.07005.

Selected Proposals

23. Charles Perin*, Tica Lin, **Lijie Yao**, Yalong Yang, Maxime Cordeil, Wesley Willett. First-Person Visualizations for Outdoor Physical Activities: Challenges and Opportunities. *Selected proposals of workshop at IEEE VIS*, Oct. 2024, Florida, United States.
24. **Lijie Yao***. Situated Visualization in Motion. *Selected proposals of Doctoral Colloquium at IEEE VIS*, Oct. 2022, Oklahoma City, United States.

Thesis

25. **Lijie Yao***. Situated Visualization in Motion. Human-Computer Interaction [cs.HC]. Université Paris-Saclay, 2023. English. NNT: 2023UPASG093. URL: theses.hal.science/tel-04413122.

RESEARCH ACTIVITIES

Grants

- Medical-Engineering Interdisciplinarity Research Grants, Ruijin Hospital | 上海交通大学医学院附属瑞金医院“青年培育计划医工交叉”项目 (Co-PI, Grant № 2026JC080) 07/2026–07/2027
- Open Project Program of the State Key Laboratory of CAD&CG, Zhejiang University | 浙江大学 CAD&CG 全国重点实验室开放课题 (PI, Grant № A2514) 01/01/2025–31/12/2026
- XJTLU RDF (PI, Grant № RDF-24-01-062) 01/01/2025–31/12/2027
- XJTLU PGRS (PI, Grant № FOSA2506030) 16/07/2025–now

Awards

- Best Paper Honorable Mention at ChinaVis July 2026
- Jiangsu Provincial Double-Innovation Doctor Program | 江苏省“双创博士” (№ JSSCBS2025067) Dec. 2025
- Best Poster Honorable Mention at ChinaVis July 2025
- ACM AFIHM Ph.D. Thesis Prize March 2024
- Canada Mitacs Globalink Research Award May. 2023

Organization

- 2026:
  Organizing Committee (Community/Meetups/Election Chair) & Program Committee (Full papers & Posters), *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 9-13, Boston, U.S.
  Program Committee (Technical papers & Art tracks), *International Symposium on Visual Information Communication and Interaction (VINCI)*, Nov. 1-4, Hangzhou, China.
  Program Committee (Full papers & Competition), *ChinaVis*, July 19-22, Guiyang, China.
- 2025:
  Program Committee (Technical papers), *International Symposium on Visual Information Communication and Interaction (VINCI)*, Dec. 1-3, Linz, Austria.
  Program Committee (Full papers & Posters), *IEEE Visualization & Visual Analytics (IEEE VIS)*, Nov. 2-7, Vienna, Austria.
  Program Committee (Short papers), *ACM Symposium on Eye Tracking Research & Applications (ETRA)*, May 26-29, Tokyo, Japan.
- 2024:
  Co-organiser for workshop on First-Person Visualizations for Outdoor Physical Activities, *IEEE visualization & Visual Analytics (IEEE VIS)*, Oct. 13, Florida, U.S.

Review

- 2026: CHI **Special Recognitions for Outstanding Reviews**, VIS, EuroVis, TVCG, PacificVis (TVCG journal track), ISMAR, ETRA, HRI, JOV, ChinaVis, VINCI, CG&A
- 2025: CHI, VIS, EuroVis, TVCG, ISMAR, UIST, ETRA, HRI, NSERC, ChinaVis, IWC, VINCI
- 2024: CHI **Special Recognitions for Outstanding Reviews**, VIS, EuroVis, TVCG, TSMC, ETRA, alt.chi, IMX, HRI, SAIS, ISS, MobileHCI, ETVIS
- 2023: CHI, VIS, EuroVis, CHI LBW
- 2022: VIS, SIGGRAPH Asia

TEACHING ACTIVITIES

Grants

- XJTLU TDF (PI, Grant № TDF2425-R30-283) 01/09/2025–31/08/2027
- XJTLU EASE (Team member, № TDFEASE-R1-2526-42) 01/09/2026–31/12/2026

Awards

- Third Prize in XIPU "AI in Education" Innovation Competition May 2026
- First-round Certificate in XJTLU Global Dream-Chasers Entrepreneurial Competition April 2026
- Excellent SURF Poster Awards × 2, Student-nominated Winner × 1 Sept. 2025
- Second Prize in XIPU "AI in Education" Innovation Competition Feb. 2025

Courses Delivery

Year	Course Name	Level	Student Numbers	Teaching Language
Xi'an Jiaotong-Liverpool University				
2026–now	CPT418 - Human-AI Interaction (Module leader)	Graduate	∼20	English
2024–now	SAT405 - FMP (MRes)	Graduate	See **Supervision** for details	English
2024–now	SAT406 - FMP (MSc)	Graduate	See **Supervision** for details	English
2024–now	CPT210 - Microprocessor system (Module leader)	Undergraduate	∼550	English
2024–now	SAT301 - FYP	Undergraduate	See **Supervision** for details	English
2024–2026	SAT006 - Foundations of Computer Science and Engineering	Undergraduate	∼720	English
Université Paris-Saclay				
2021–2023	Interactive Information Visualization	Graduate	∼50	English

Supervision

Year	Degree-granting Institution	Program	Student (F/M)	Outcomes	Next Step
Post-graduate					
2026–now	University of Liverpool	PhD	Junhao Zhao	1 × TVCG full paper 1 × IEEE VIS poster	
Graduate					
2025–2026	University of Liverpool	MSc	Yiming Zhang		Entrepreneurship
2025–2026	University of Liverpool	MSc	Shiyuan Jiang		Public-sector
2025–2026	University of Liverpool	MSc	Zehong Ye	1 × ChinaVis full paper	NUVERSE Games
2025–2026	University of Liverpool	MSc	Yihan Dong (F)		BOCHU Electronic
2025–2026	University of Liverpool	MSc	Fanhua Liu		AMD
2024–2026	University of Liverpool	MRes	Yuzhu Chen (F)		BOCOM

Year	Degree-granting Institution	Program	Student (F/M)	Outcomes	Next Step
2021–2022	Université Paris-Saclay	MSc	Federica Bucchieri (F)	1 × TVCG full paper 1 × EuroVis poster 1 × JournéeVisu poster	3DClouds
Undergraduate					
2025–2026	XJTLU & University of Liverpool	BEng	Haochen Yan	1 × Excellent SURF Award 1 × Student-nominated Winner	Master at UCL
2025–2026	XJTLU & University of Liverpool	BEng	Ziyuan Wang	1 × Excellent SURF Award	Master at UoE
2025–2026	XJTLU & University of Liverpool	BEng	Weixin Xiong (F)	1 × Excellent SURF Award	Master at NUS
2025–2026	XJTLU & University of Liverpool	BEng	Yating Wu (F)		Master at UCL
2025–2026	XJTLU & University of Liverpool	BEng	Xiaoran Zhao (F)		Master at UoM
2025–2026	XJTLU & University of Liverpool	BEng	Xiaoyu Li (F)		Master at UoB
2025–2026	XJTLU & University of Liverpool	BSc	Ruoxi Liao (F)	1 × Excellent SURF Award 1 × Student-nominated Winner	Master at NUS
2025–2026	XJTLU & University of Liverpool	BSc	Tiantian Ye (F)		Master at UCL
2025–2026	XJTLU & University of Liverpool	BSc	Xi Zhou (F)	1 × IEEE VIS poster	Master at UoE
2024–2025	XJTLU & University of Liverpool	BEng	Xing Wei		Master at USYD
2024–2025	XJTLU & University of Liverpool	BEng	Yuming Zhang	1 × IEEE VIS workshop paper	Work in industry
2024–2025	XJTLU & University of Liverpool	BEng	Yongxuan Teng		Meituan
2024–2025	XJTLU & University of Liverpool	BEng	Dingkang Yang		Master at Monash
Summer Undergraduate Research Fellowship (SURF)					
2025		SURF	Qi Guo (F)	1 × IEEE VIS workshop paper 1 × Excellent SURF Award 1 × Student-nominated Winner	
2025		SURF	Junyi Li (F)	1 × IEEE VIS workshop paper 1 × Excellent SURF Award 1 × Student-nominated Winner	
2025		SURF	Ziyi Jin (F)	1 × Excellent SURF Award	
2025		SURF	Xun Gong (F)	1 × Excellent SURF Award	
2025–now		SURF	Jingyu Wang (F)	1 × Excellent SURF Award 1 × Student-nominated Winner 2 × ChinaVis full paper 1 × ChinaVis Best Paper Honorable Mention	
Research Assistants					
2025–2026			Junhao Zhao	1 × IEEE VIS workshop paper	PhD at XJTLU
2024–2025			Haonan Yao	1 × IEEE VIS short paper	PhD at UNNC